

Winning baseball games by solving statistics puzzles

Scott Powers

TAMIDS Seminar Series
February 16, 2026

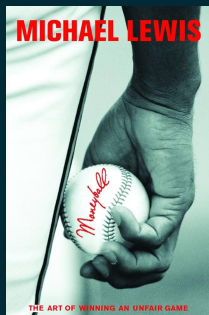


RICE UNIVERSITY
Sport Analytics

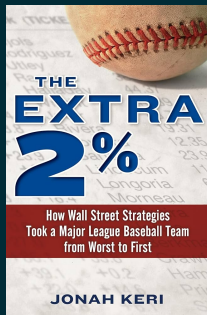
What is Sport Analytics?



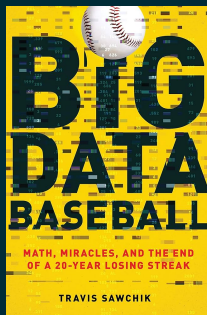
Books about Baseball Teams Using Analytics to Win



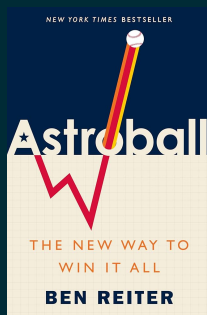
2002
Oakland
Athletics



2008
Tampa Bay
Rays



2013
Pittsburgh
Pirates



2017
Houston
Astros

MLB R&D Staff Sizes (according to Reddit)

Team	Analysts	Team	Analysts
American League		National League	
Yankees	20	Dodgers	20
Astros	15	Braves	15
Rays	15	Brewers	11
Angels	10	Reds	11
Tigers	9	Phillies	10
Rangers	8	Nationals	8
Mariners	7	Pirates	8
Royals	7	Padres	7
Twins	7	Cardinals	6
Blue Jays	6	Cubs	6
Red Sox	6	Giants	6
Indians	5	Marlins	6
Orioles	5	Dbacks	5
Athletics	3	Rockies	4
White Sox	2	Mets	3

2018

MLB Teams Ranked by Approximate R&D Size			
Includes Data Scientists/Analysts, Data/Software Engineers, Biomechanists, etc.			
TEAM	SIZE	TEAM	SIZE
	42		20
	41		20
	36		20
	35		20
	34		19
	34		19
	29		18
	28		17
	27		16
	26		16
	25		16
	24		15
	24		12
	23		12
	20		12
	20		10

Teams With No Clear Info: A's, Angels, Cubs, D-Backs

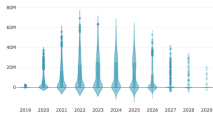
Amrit Vignesh | @avsportsanalyst

2025

What are they all doing?

Player Evaluation

Executives use modeling results to decide which players to target in trades and in free agency.



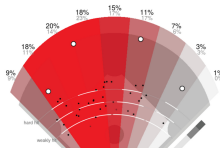
Player Development

Players use data-driven insights to tweak their game and achieve greater levels of performance.



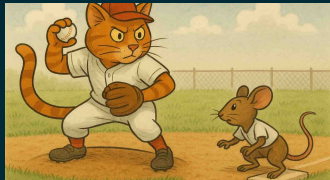
In-Game Strategy

Coaches use analytics tools to position fielders and decide when to make substitutions.



Three Puzzles

I: How far should you stray from first base?



II: Should you just swing as hard as you can?

HOW LONG IS YOUR SWING?

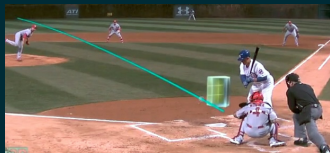
A SWING'S LENGTH IS CAPTURED FROM THE START OF A SWING UNTIL IMPACT POINT.

A SWING CAN BE AS SHORT AS 4 FEET OR AS LONG AS NEARLY 10 FEET.

A white line drawing on a dark blue background showing a baseball player in mid-swing. A curved arrow indicates the path of the bat from the start of the swing to the impact point.

STATCAST powered by Google Cloud

III: What does a pitch tell you about a pitcher?



How far should you stray from first base?



Sivaramakrishnan Ramani, Rice University



Jacob Hahn, Rice University → Chicago Cubs



Andrew Schaefer, Rice University

Powers S, Ramani S, Hahn J, Schaefer AJ (2026)

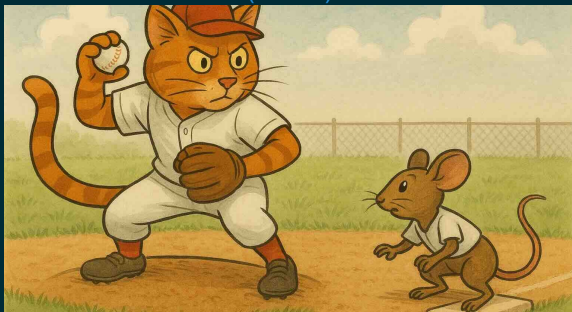
“Lead distance under a pickoff limit in Major League Baseball: A sequential game model”

arXiv:2601.15608

Background

In 2023, MLB limited the pitcher to **3 pickoff attempts**

(video)



After **3 failed pickoffs**, the runner advances automatically

The Puzzle

How should the runner choose their lead distance to maximize **expected runs** scored in the inning?

We got 2022–2023 **pitch-by-pitch lead distance** from MLB*

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Step 1: Estimate the effect of lead distance

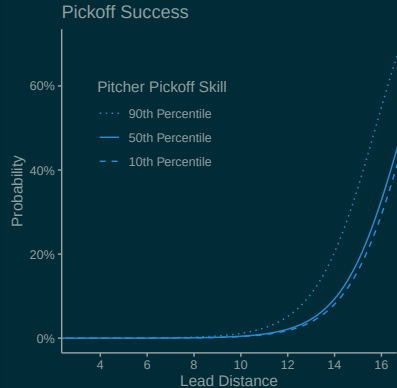
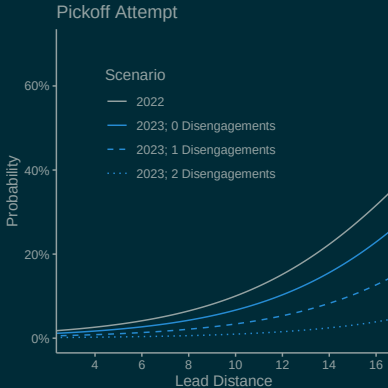
How does lead distance influence...

- **Pickoff attempt** probability?
- **Pickoff success** probability?
- **Stolen base success** probability?

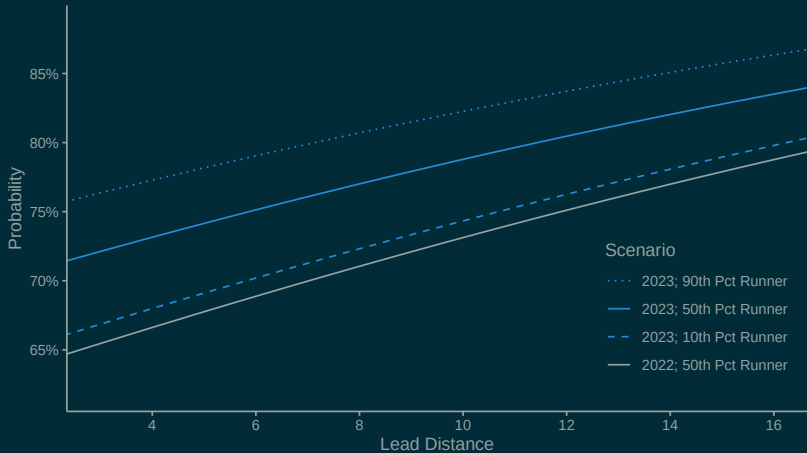
generalized linear mixed-effects model (GLMM)

- binomial likelihood for each outcome
- fixed effects: lead distance, disengagements, out, count, runner sprint speed, catcher arm strength
- random effects: pitcher, runner, catcher

Pickoff attempts increase with lead distance



Stolen bases increase with lead distance



Step 2: Solve for the optimal lead distance policy

What is the **game-theoretic equilibrium** strategy?

two-player sequential game

- state space: bases occupied, outs, count, disengagements
- action space: lead distance (runner), pickoff (pitcher)
- reward function: runs scored
- solved via **value iteration**

The Two-Foot Rule

Count	Disengagements		
	0	1	2
0-0	10.0	11.5	14.1
1-0	10.2	11.7	14.2
2-0	10.2	11.7	14.2
3-0	10.3	11.8	14.2
0-1	9.8	11.4	14.1
1-1	10.0	11.6	14.2
2-1	10.3	11.8	14.3
3-1	10.4	11.9	14.3
0-2	9.5	11.1	14.1
1-2	9.9	11.4	14.3
2-2	10.0	11.5	14.2
3-2	10.4	11.9	14.4

- Increase lead by **two feet** after each disengagement
- Value of implementation:
~12 runs / team / season
(~\$12,000,000)

Should you just swing as hard as you can?



Ron Yurko, Carnegie Mellon University

Powers S, Yurko R (2026) “Swinging, Fast and Slow: Interpreting variation in baseball swing tracking metrics”
arXiv:2507.01238, to appear in *The American Statistician*

Background

Fans lament batters not having **two-strike approaches**

(video)



HOW LONG IS YOUR SWING?

A SWING'S LENGTH IS CAPTURED FROM THE START OF A SWING UNTIL IMPACT POINT.

A SWING CAN BE AS SHORT AS 4 FEET OR AS LONG AS NEARLY 10 FEET.

The infographic features a white line-art illustration of a batter in mid-swing on a dark blue background. A white curved arrow traces the path of the bat from the start of the swing to the impact point with the ball. The entire graphic is set against a dark blue background with a light blue grid pattern.

STATCAST POWERED BY Google Cloud

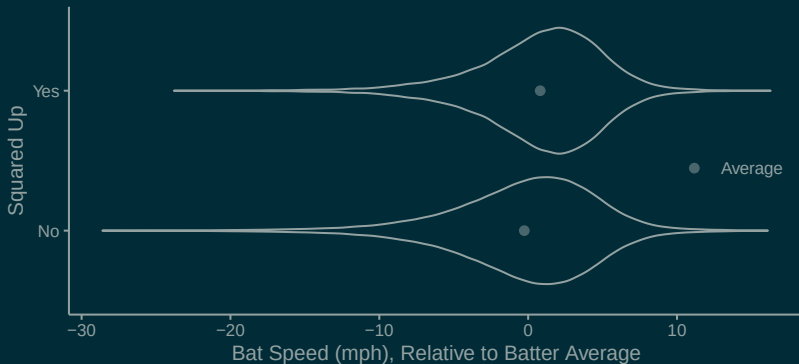
MLB logo

savant

May 2024: MLB releases **swing length** and **bat speed** data

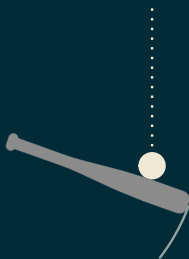
Faster swings make better contact?

"Squared Up" = good contact (exit velocity > 80% of theoretical max)

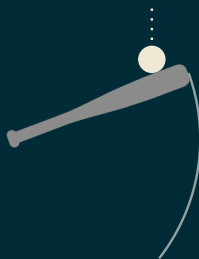


Key Insight

We can't take swing metrics at **face value**



LATE Swing



EARLY Swing

Baseball Savant reports swing metrics at **point of contact**

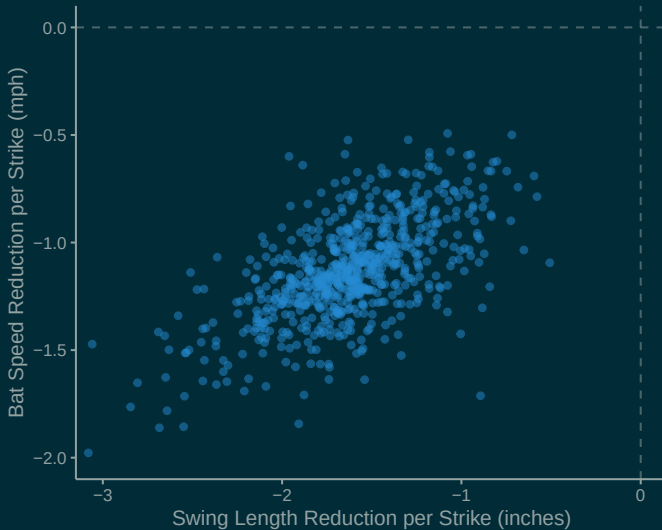
Step 1: Infer batter intention

What does the batter's swing look like
when they make solid contact?

Bayesian hierarchical model

- skew-normal likelihood for bat speed and swing length
- fixed effects: count, location
- random effects: batter, batter $\times$ count, batter $\times$ location

Some batters shorten/slow their swings more



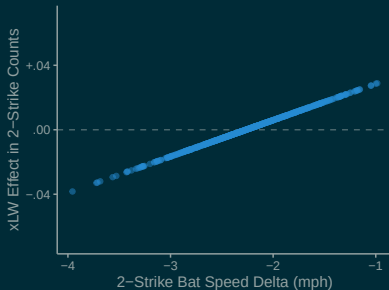
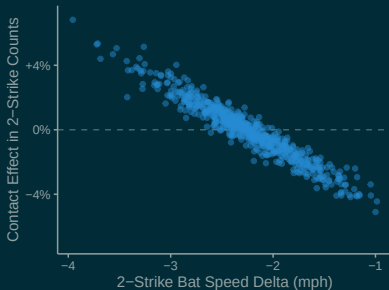
Step 2: Estimate effect of intention

If you make your swing **shorter/slower** with two strikes, do you make **more contact** with two strikes?

Instrumental variables regression (causal inference)

- logistic regression for contact, linear regression for power
- instrument: count
- treatment: batter $\times$ count random effect
- control for difficulty of pitch (gradient boosting model)

Slower swings mean more contact, less power



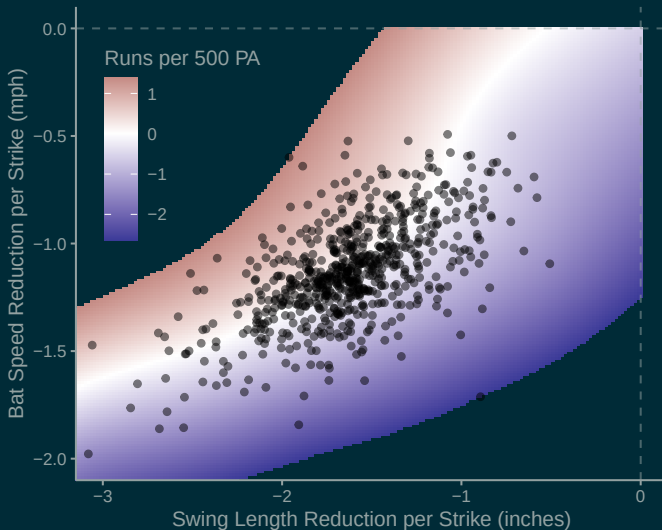
Step 3: Value the contact-power tradeoff

Is **increasing** your contact rate worth **decreasing** your power?

Markov chain model

- state space: ball-strike count
- transition probabilities: depend on approach
- value function: linear weight of plate appearance outcome

The tradeoff is (roughly) balanced



What does a pitch tell you about a pitcher?



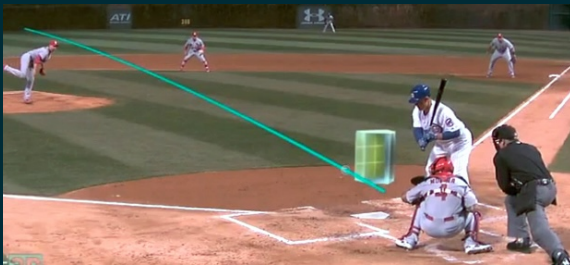
Vicente Iglesias, Susquehanna International Group

Working paper:

Powers S, Iglesias V “Pitch trajectory density estimation for predicting future outcomes”

Background

Full **pitch tracking** data have been available for 20 years

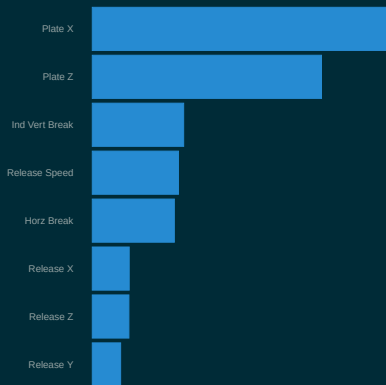


Teams evaluate pitchers based on **pitches**, not outcomes

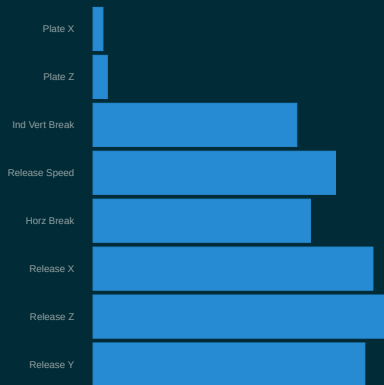
(video #1) (video #2)

Evaluating the *pitch* $\neq$ evaluating the *pitcher*

Variable Importance¹



Variable Reliability²



¹ fractional contribution of each feature's splits to gradient boosting pitch model

² (between-pitcher variance) / (total variance); varies by pitch type (here: RHB FB)

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
...		
x_n	$\rightarrow$	y_n

$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

Our Problem

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
	...		
C	x_n	$\rightarrow$	y_n



$$A \rightarrow \hat{y}$$

Key Insight

Supervised Learning

Pitch		Outcome
x_1	$\rightarrow$	y_1
x_2	$\rightarrow$	y_2
x_3	$\rightarrow$	y_3
$\dots$		
x_n	$\rightarrow$	y_n

$$x^* \rightarrow \hat{y} = \hat{f}(x)$$

Our Approach

Pitcher	Pitch		Outcome
A	x_1	$\rightarrow$	y_1
A	x_2	$\rightarrow$	y_2
B	x_3	$\rightarrow$	y_3
	$\dots$		
C	x_n	$\rightarrow$	y_n



$$A \quad \hat{p}(x|A) \rightarrow \int \hat{p}(x) \hat{f}(x)$$

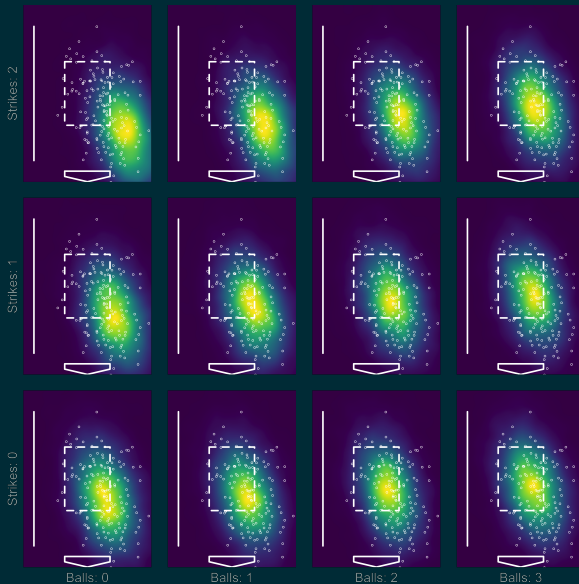
Step 1: Estimate distributions over pitch trajectories

What is the **generative model** for pitch trajectories?

Bayesian hierarchical model

- multivariate normal likelihood (9 dimensions)
- mean $\sim$ pitcher, context, pitcher $\times$ context
- variance scaling $\sim$ pitcher, context
- correlation $\sim$ pitcher

Dylan Cease's Slider vs RHB in All Counts



Step 2: Predict outcomes of pitch trajectories

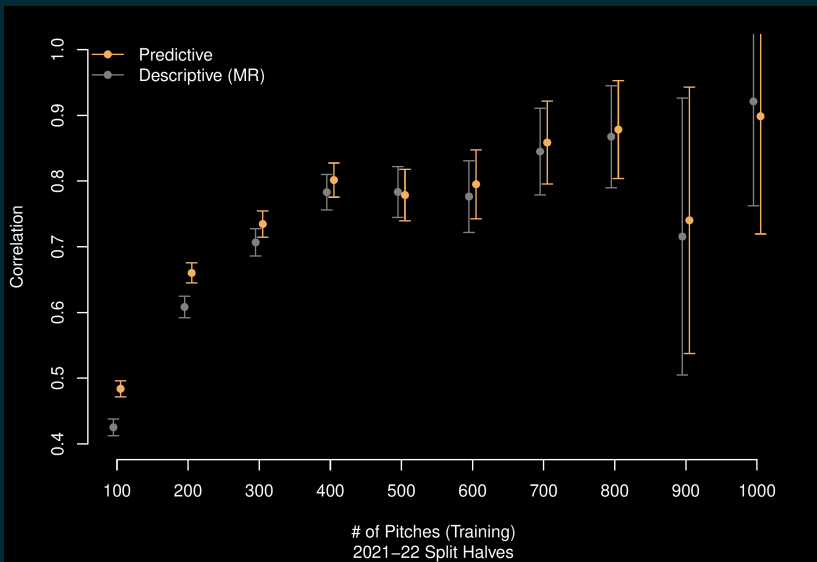
What is the **run value** $\hat{f}(x)$ of each possible trajectory?

gradient boosting

- predict run value of pitch

$$\text{predictive pitch score} = \int \left\{ \hat{p}(x|\text{pitcher, context}) \cdot \hat{f}(x) \right\} dx$$

Predictive pitch score wins for < 300 pitches



Thank You!

saberpowers.github.io/research

Powers S, Ramani S, Hahn J, Schaefer AJ (2026)

“Lead distance under a pickoff limit in Major League Baseball: A sequential game model”

arXiv:2601.15608

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